

Anthic Kumar Singh

Software Engineer & Data Scientist

Kuril Bishwa Road, Dhaka, Bangladesh | +880 1717182035, +880 1779080742

anthickumarsingh2@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)



Professional Summary

Software engineer and data scientist specializing in AI-driven applications, machine learning systems, and scalable full-stack architectures. Experienced in building production-grade AI agents, retrieval-augmented generation (RAG) systems, predictive analytics platforms, and cloud-deployed web applications. Passionate about leveraging data, automation, and generative AI to solve real-world problems and deliver measurable business impact.

Technical Skills

Languages & Frameworks: JavaScript (ES6+), TypeScript, Python, React, Redux, Next.js, Node.js, Express.js, Django, FastAPI, R, SPSS, Stata
Machine Learning & Data: Scikit-learn, Pandas, NumPy, XGBoost, SHAP, LangGraph, LangChain, RAG, LLM, Agentic AI, Generative AI
Databases & Cloud: MongoDB, Redis, Qdrant, Supabase (PostgreSQL), Firebase, Vercel, Docker
Tools & Methodologies: Git, Nginx, Streamlit, Jest, SSE, JWT, CI/CD, Webpack, Tailwind CSS, HTML5, CSS3

Professional Experience

Full Stack Developer **February 2025—Present**
The Nexgenix, Chittagong, Bangladesh | [thenexgenix.com](#)

- Developed responsive web applications using React.js, TypeScript, and Tailwind CSS, achieving cross-device compatibility across 10+ screen sizes
- Architected dynamic UI components and optimized rendering performance, delivering measurable UX improvements
- Integrated REST APIs via Axios, reducing data-fetch latency and streamlining client-server communication
- Deployed full-stack applications on Linux VPS servers, configuring Node.js/Express.js services, SSL, domains, and production builds
- Managed application state with Redux Toolkit and Context API; implemented CI/CD pipelines for seamless deployment workflows

Projects

Atlas.AI — Production-Grade Multi-Agent Research System **2026**
[Live Demo](#) | [Frontend](#) | [Backend](#) | [Agent Engine](#)

- Architected a 9-node LangGraph multi-agent pipeline achieving >92% average critique score with a self-correcting Critic Agent rewrite loop (score < 8/10), backed by a 4-provider AI fallback cascade (Mistral → Groq → Gemini → OpenRouter) reducing zero-output failures to <0.5%.
- Engineered a Node.js/Express TypeScript API gateway with dual-token JWT auth (15-min access + 7-day refresh rotation), Upstash Redis caching serving repeated queries in <50ms vs 90+ second pipeline runs, and an async SSE job queue eliminating HTTP gateway timeouts across 8 route modules
- Built 6 specialized academic AI Agents (Literature Gap Finder, 3-Agent Peer Review Panel, Thesis Defense Simulator, Temporal Fact-Checker, Slide Generator, and AI Paraphraser with 4 modes) with Qdrant vector vault RAG, improving factual accuracy 38% vs non-RAG baseline
- Deployed across a 6-service multi-cloud stack: Vercel (frontend + gateway), AWS EC2 (AI engine), Qdrant Cloud (RAG), Supabase PostgreSQL (history), Upstash Redis (cache), MongoDB Atlas (users/payments) — achieving 85.7% pass rate across 28 automated end-to-end feature tests

COVID-19 Prediction & Reporting Web App **2025**
[Live Demo](#) | [Frontend](#) | [Backend](#)

- Spearheaded the design and implementation of a production-grade 9-node LangGraph multi-agent architecture that autonomously performs query expansion, web intelligence gathering, RAG-based knowledge retrieval, fact verification, and self-correcting report generation using Mistral models, reducing manual research effort by over 90%
- Engineered a scalable TypeScript API gateway with JWT authentication, refresh-token rotation, Redis-backed rate limiting, asynchronous job orchestration, CSRF protection, and role-based access control, eliminating long-running request bottlenecks and ensuring secure AI workload execution
- Developed an interactive Next.js 15 and React 19 research workspace featuring real-time Server-Sent Events (SSE), custom LangGraph workflow visualization, live agent execution tracking, automated report scoring, and historical research analytics dashboards
- Orchestrated a distributed cloud-native deployment architecture across Vercel, Google Cloud Run, Qdrant Cloud, Supabase PostgreSQL, Upstash Redis, and MongoDB Atlas, enabling cost-efficient, highly available AI research workflows on a fully serverless infrastructure

Education

B.Sc. in Statistics **CGPA: 3.27/4.00**
Mawlana Bhashani Science and Technology University, Tangail, Bangladesh

Relevant Coursework: Statistical Analysis, Probability, Data Science, Mathematical Modeling, Research Methods, Computational Statistics

Leadership, Certifications & Training

- Volunteer Coordinator—Programming Hero Reunion Programme** [Certificate](#)
Organized and coordinated technical workshops and community events for 200+ participants, supporting event operations and engagement activities.
- Complete Research Training: From Zero to Publication** [Certificate](#)
Bangladesh Advanced Research Training Center (BARTC)
Completed intensive training in research methodology, academic writing, literature review, data analysis, and the scientific publication process.
- Database (MySQL/Oracle/SQL) Training** [Certificate](#)
Enhancing Digital Government and Economy (EDGE) Project, Bangladesh Computer Council (BCC), ICT Division
Hands-on training in DBMS, SQL, MySQL, Oracle Database, schema design, querying, and database administration.